DataFlix: A Hybrid Matrix Factorization Recommendation System for Netflix Movie Ratings

Aryan Kumar [1] **Sura Sravan Kumar** [1] **Shreyash Pandit** [1]

[1] *Department of Computer Science & Engineering, Indian Institute of Technology Gandhinagar*

{aryan.kumar, sravan.sura, shreyash.pandit}@iitgn.ac.in

Abstract. Recommender systems are central to modern streaming platforms, yet collaborative filtering methods often struggle with the sparsity inherent in large-scale rating datasets. This proposal describes *DataFlix*, a hybrid recommendation system that employs Matrix Factorization (MF) with incomplete data as its core method, evaluated on the Netflix Movie Rating Dataset. We train two complementary model variants: a regression model optimised with Mean Squared Error (MSE) for accurate rating prediction, and a ranking model optimised with Bayesian Personalised Ranking (BPR) for top-K recommendation quality. Both variants fuse low-rank MF latent factors with Sentence-BERT content embeddings and metadata features via a projected, self-attention fusion layer. The Netflix dataset is augmented with IMDb and TMDb side-information. Training uses ALS and Adam-based gradient descent with ℓ_2 regularisation. Evaluation on held-out Netflix ratings covers RMSE, MAE, MRR, Precision@K, Recall@K, NDCG@K, and Intra-List Diversity, benchmarked against five baselines with full ablation studies and a dedicated cold-start protocol.

CCS Concepts: Information systems → *Recommender systems*; *Collaborative filtering*; Computing methodologies → *Neural networks*.

Keywords: Recommender Systems, Matrix Factorization, BPR, Self-Attention Fusion, Sentence-BERT, Netflix Dataset, ALS, Adam, Cold-Start.

1. Introduction and Motivation

Streaming services such as Netflix, Amazon Prime, and Jio Hotstar must surface relevant content from catalogues of thousands of titles to millions of users. A well-calibrated recommendation engine can increase user engagement, reduce churn, and maximise content discovery.

The Netflix Prize Competition (2006–2009) demonstrated that even a 1% improvement in rating-prediction accuracy can have significant commercial value [?]. The core challenge is that real-world rating matrices are *extremely sparse* ($\approx 1.2\%$ density in the Netflix corpus): each user rates only a tiny fraction of available titles. Moreover, the *absence* of a rating carries its own preference signal, motivating implicit-feedback models.

Classical user-based or item-based collaborative filtering suffers from scalability and cold-start issues. Matrix Factorization [?] addresses this with compact latent representations; SVD++ [?] extended MF to incorporate implicit item sets. Deep neural approaches such as NeuMF [?] replace the inner product with a non-linear interaction, but discard rich textual and metadata signals entirely.

DataFlix addresses these gaps through a hybrid framework that (i) separates the rating-prediction and ranking objectives cleanly into two training paths, (ii) fuses MF latent factors with Sentence-BERT content embeddings via a projected self-attention layer, and (iii) improves cold-start coverage using interaction-weighted content profiles. The project goals are:

1. Train ALS and Adam-based MF models on the sparse Netflix matrix with bias terms and ℓ_2 regularisation.
2. Augment movie embeddings with learnable genre projections and SBERT [?] text representations.
3. Augment user embeddings with rating-weighted interaction history.
4. Train a BPR ranking variant for top-K recommendation.
5. Project all feature streams to a common dimension and fuse via self-attention before the prediction head.
6. Evaluate comprehensively against five baselines including a strong cold-start SBERT cosine-similarity baseline.

2. Related Work

Collaborative Filtering and MF. Koren et al. [?] established MF as the dominant CF paradigm. SVD++ [?] enriched user vectors with implicit item sets. Hu et al. [?] proposed WRMF for purely implicit feedback (e.g. click counts), assigning log-scaled confidence weights to unobserved interactions.

Ranking-Aware Optimisation. Rendle et al. [?]

introduced BPR, a pairwise AUC-consistent loss widely adopted for implicit feedback recommenders. BPR consistently outperforms pointwise MSE-trained models on top-K ranking metrics.

Deep and Hybrid Models. He et al. [?] proposed NeuMF, fusing GMF and MLP pathways. Zhang et al. [?] survey hybrid approaches that augment latent factors with content embeddings for strong cold-start gains. DataFlix extends this line with a self-attention fusion layer and a BPR-trained ranking path.

Text Embeddings in RS. Reimers and Gurevych [?] showed that Sentence-BERT produces semantically rich, fixed-size representations that transfer well across tasks. SBERT synopsis embeddings yield movie representations that generalise to unseen items in a way that sparse one-hot genre vectors cannot.

3. Dataset Description

3.1 Evaluation Dataset: Netflix Movie Ratings

The Netflix Movie Rating Dataset (Kaggle[1]) is derived from the original Netflix Prize corpus and serves as the *primary evaluation benchmark*. It contains **over 17 million explicit ratings** from approximately **480,000 users** across **17,770 movies**, on a 1–5 integer scale, with an overall matrix density of $\approx 1.2\%$.

Key fields:

- `MovieID` — unique integer identifier.
- `CustomerID` — anonymised user identifier.
- `Rating` — integer $\in \{1, 2, 3, 4, 5\}$.
- `Date` — ISO timestamp of the rating event.

3.2 Augmentation Sources

- **IMDb / TMDb metadata** — genre tags, director, cast, runtime, language, and critic scores, joined on title and release year.
- **Plot synopses** — retrieved via the TMDb API and encoded with SBERT for semantic embeddings.
- **MovieLens 25M** [?] — used for genre taxonomy cross-validation and to supplement cold-start user profiles absent from the Netflix training split.

3.3 Pre-processing Pipeline

1. Remove users with < 5 ratings and movies with < 10 ratings.
2. Subtract the per-user mean rating (user bias correction).
3. Apply a *global* temporal split at the 80th-percentile date so all users share the same train/validation/test cutoff, preventing leakage from future ratings. Approximately 12,000 users have < 3 training interactions and form the cold-start evaluation split.
4. For the BPR ranking path, binarise ratings: interactions with normalised score > 0 are treated as positives; all unobserved items are candidate negatives.
5. Store the training split as a sparse CSR matrix for GPU-accelerated factorisation.

[1] `https://www.kaggle.com/datasets/rishitjavia/netflix-movie-rating-dataset`

4. Methodology

4.1 Problem Formulation

Let $M \in \mathbb{R}^{U\times I}$ be the partially observed rating matrix with observed index set Ω. We seek a low-rank factorisation:

$$\hat{M} = PQ^\top, \quad P \in \mathbb{R}^{U\times k},\ Q \in \mathbb{R}^{I\times k}, \tag{1}$$

where $k \in \{50, 100, 200\}$ is the latent dimension. The biased prediction is:

$$\hat{r}_{ui} = \mu + b_u + b_i + \mathbf{p}_u^\top \mathbf{q}_i. \tag{2}$$

4.2 Path A: Rating Regression (MSE)

We minimise the standard regularised squared error over *observed* entries only (masked reconstruction):

$$\mathcal{L}_{\text{MSE}} = \sum_{(u,i)\in\Omega} (r_{ui} - \hat{r}_{ui})^2 + \lambda(\|\mathbf{p}_u\|^2 + \|\mathbf{q}_i\|^2 + b_u^2 + b_i^2). \tag{3}$$

Gradients are masked to Ω; unobserved entries receive no gradient signal (no WRMF count-based weighting is needed since Netflix provides explicit ratings, not interaction counts).

4.2.1 ALS.

ALS alternates closed-form updates fixing Q, solving for P, then vice versa. For user u:

$$\mathbf{p}_u = \left(\sum_{i\in\Omega_u} \mathbf{q}_i\mathbf{q}_i^\top + \lambda\mathbf{I}\right)^{-1} \sum_{i\in\Omega_u} r_{ui}\,\mathbf{q}_i. \tag{4}$$

ALS is trivially parallelisable and avoids learning-rate tuning. Convergence is declared when the relative decrease in $\mathcal{L}_{\text{MSE}}$ is $< 10^{-4}$ over one complete sweep.

4.2.2 Adam-Based Gradient Descent.

We also train using the Adam optimiser [?] (not vanilla SGD) with cosine-annealing learning-rate decay ($\eta_0 = 5\times10^{-3}$, $T_{\max} = 50$ epochs) and early stopping on validation RMSE with patience of 5 epochs.

4.3 Path B: Ranking Optimisation (BPR)

For top-K recommendation we train a separate model variant with the Bayesian Personalised Ranking loss [?]. Using the binarised implicit signal (Section 3.3), for each user u and observed positive item i, a negative item j is sampled from unobserved items with popularity-proportional probability (harder negatives):

$$\mathcal{L}_{\text{BPR}} = -\sum_{(u,i,j)\in\mathcal{D}_S} \ln\sigma(\hat{r}_{ui}-\hat{r}_{uj})+\lambda(\|\mathbf{p}_u\|^2+\|\mathbf{q}_i\|^2+\|\mathbf{q}_j\|^2), \quad (5)$$

where σ is the sigmoid. Path B is trained exclusively with $\mathcal{L}_{\text{BPR}}$ and evaluated solely on ranking metrics (NDCG@K, MRR, Precision@K, Recall@K). Path A is evaluated on rating metrics (RMSE, MAE). The two paths share the same embedding architecture but are trained independently, avoiding objective conflicts.

5. Embeddings and Architecture

5.1 Feature Streams

All feature vectors are first projected to a common dimension $d = 128$ before fusion to ensure dimensional compatibility in the attention layer.

Movie streams (i):

1. $\mathbf{q}_i^{(d)} = W_q\,\mathbf{q}_i$, projection of the MF latent vector $\mathbf{q}_i \in \mathbb{R}^k$.
2. $\mathbf{g}_i^{(d)}$: genre embedding from a learnable table (nn.Embedding over 20 genres), summed and projected to $\mathbb{R}^d$.
3. $\mathbf{t}_i^{(d)} = W_t\,\mathbf{t}_i^{\text{SBERT}}$, learnable linear projection of the SBERT [?] synopsis embedding ($\mathbb{R}^{768} \to \mathbb{R}^d$); no PCA is needed since the projection is trained end-to-end.
4. $\mathbf{s}_i^{(d)} = W_s\,[s_i]$, projection of the log-normalised popularity scalar.

User streams (u):

1. $\mathbf{p}_u^{(d)} = W_p\,\mathbf{p}_u$, projection of $\mathbf{p}_u \in \mathbb{R}^k$.
2. $\mathbf{h}_u^{(d)} = W_h\,\bar{\mathbf{t}}_u^{\text{SBERT}}$, projection of the rating-weighted mean SBERT vector: $\bar{\mathbf{t}}_u = \frac{\sum_{i\in\mathcal{H}_u}(r_{ui}-\bar{r}_u)\,\mathbf{t}_i^{\text{SBERT}}}{\sum_{i\in\mathcal{H}_u}|r_{ui}-\bar{r}_u|}$, where $\mathcal{H}_u$ is u's rating history.
3. $\mathbf{f}_u^{(d)} = W_f\,\mathbf{f}_u$, projection of the rating-pattern feature vector $\mathbf{f}_u$: mean rating, rating std. dev., log-count of rated items, and fraction of ratings above user mean.

5.2 Self-Attention Fusion

After projection all streams share dimension d, enabling multi-head self-attention. Let $X_i \in \mathbb{R}^{4\times d}$ be the row-stacked item stream matrix (and $X_u \in \mathbb{R}^{3\times d}$ for users). We apply one-layer multi-head self-attention:

$$\text{Attn}(X) = \text{softmax}\left(\frac{XW_Q(XW_K)^\top}{\sqrt{d_h}}\right)XW_V, \quad (6)$$

where $W_Q, W_K, W_V \in \mathbb{R}^{d\times d_h}$ are learned projections with $d_h = d/H$ per head ($H = 4$ heads). The fused embedding $\mathbf{e}_i$ is obtained by mean-pooling the attended rows: $\mathbf{e}_i = \frac{1}{4}\sum_m[\text{Attn}(X_i)]_m$. This formulation is input-dependent (unlike a global query vector), allowing each movie or user to weight its own streams differently.

5.3 Architecture Overview

Figure 1 shows the full DataFlix pipeline.

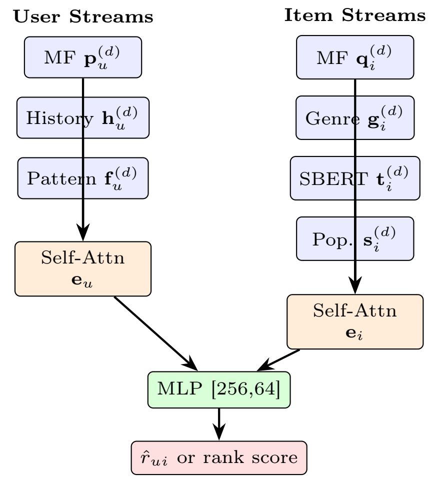

Figure 1: DataFlix architecture. All feature streams are projected to $\mathbb{R}^d$, fused via multi-head self-attention, then passed to a shallow MLP. Path A outputs a clipped rating; Path B outputs a ranking score.

5.4 Prediction Head

$$\hat{r}_{ui} = \text{MLP}(\mathbf{e}_u \,\|\, \mathbf{e}_i), \quad (7)$$

with hidden layers [256, 64], ReLU activations, dropout ($p = 0.2$). Path A clips the output to $[1, 5]$; Path B applies sigmoid to yield a preference score.

5.5 Implementation Details

- **Framework**: PyTorch 2.x with `torch.sparse` for memory-efficient sparse matrix operations.
- **Hardware**: NVIDIA RTX-series GPU; ALS subproblems solved in parallel via `scipy.sparse.linalg.lsqr`.
- **Hyperparameter tuning**: Optuna [?] TPE search over $k \in \{50, 100, 200\}$, $\lambda \in [10^{-4}, 10^{-1}]$, $\eta_0 \in [10^{-4}, 10^{-2}]$, $H \in \{2, 4, 8\}$, $d \in \{64, 128, 256\}$.

6. Evaluation Plan

6.1 Metrics

Path A (Rating Prediction):

- **RMSE** and **MAE** on the held-out Netflix test split.

Path B (Top-K Ranking):

- **Precision@K**, **Recall@K** ($K \in \{5, 10, 20\}$).
- **NDCG@K** — position-weighted ranking quality.

- **MRR** — Mean Reciprocal Rank of the first relevant item.
- **ILD@10** — Intra-List Diversity (mean pairwise SBERT cosine dissimilarity of recommended items), measuring beyond-accuracy quality.
- **Coverage** — fraction of catalogue recommended to ≥ 1 user.

6.2 Baselines

1. **Global mean** — $\hat{r}_{ui} = \mu$.
2. **Bias-only** — $\hat{r}_{ui} = \mu + b_u + b_i$.
3. **User-KNN CF** — cosine similarity, $k = 50$ neighbours.
4. **Vanilla MF (SGD)** — plain matrix factorisation with no bias terms or content features.
5. **SVD++ [?]** — MF with implicit item sets; stronger than vanilla MF.
6. **NeuMF** [?] — neural CF baseline.

6.3 Expected Results

Table 1 shows the planned evaluation grid.

Table 1: Planned evaluation grid (results to be filled in). †Path B only. ‡Path A only.

Model	RMSE‡	NDCG@10†	MRR†	ILD†
Global Mean	—	—	—	—
Bias-Only	—	—	—	—
User-KNN CF	—	—	—	—
Vanilla MF	—	—	—	—
SVD++	—	—	—	—
NeuMF	—	—	—	—
MF only	—	—	—	—
MF + Genre	—	—	—	—
MF + SBERT	—	—	—	—
MF + BPR	—	—	—	—
MF + Attn	—	—	—	—
DataFlix (full)	—	—	—	—

6.4 Ablation Studies

- MF only (no content, no BPR, no attention).
- MF + genre embeddings.
- MF + SBERT text embeddings.
- MF + BPR (no content).
- MF + content, concat fusion (no attention).
- MF + content, self-attention fusion (no BPR).
- Full DataFlix — Path A (MSE + attention + content).
- Full DataFlix — Path B (BPR + attention + content).

6.5 Cold-Start Evaluation

Users with < 3 training ratings ($\approx$12,000, 0.6% of interactions) form the cold-start split. We report NDCG@10 and Recall@10 for three cold-start settings:

1. **Content-only** (SBERT embeddings, no MF factors).
2. **SBERT cosine similarity** — zero-shot baseline: recommend K movies most similar to the user's few rated items by cosine distance in SBERT space.
3. **DataFlix** — uses content embeddings in place of absent MF vectors (set $\mathbf{p}_u = \mathbf{0}$; rely on $\mathbf{h}_u$ from the few rated items).

Acknowledgments

This work is carried out as part of **CS 328: Introduction to Data Science**, IIT Gandhinagar, 2026. The authors thank the course instructor and TAs for guidance on project scope and evaluation design.

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